Imperial College London

BSc/MSci EXAMINATION May 2020

This paper is also taken for the relevant Examination for the Associateship

PRINCIPLES OF INSTRUMENTATION

For 3rd/4th-Year Physics Students

Thursday, 14th May 2020: 10:00 to 12:30

The paper consists of two sections: A & B.
Section A contains one question, comprising of small parts. [40 marks total]
Section B contains four questions on selected parts of the course. [30 marks each]
A table of Laplace transformations is included at the end of the paper.
Candidates are required to:
Answer **ALL** *parts of Section A and* **TWO** *questions from Section B.*
Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

At the top of each page of your answers, write your CID number, module code, question number and page number. Scan and upload your answers to the Turnitin dropboxes as described in the guidance documents in the Blackboard module for this exam. Upload each answer to the dropbox provided for that specific question.

Your uploaded file name should be of the form CID_ModuleCode_QuestionNumber(s).pdf

For each answer you should prepare a coversheet which should be the first page of your scanned answer. The coversheet should contain the following:

- your CID
- module name and code
- the question number
- the number of pages in your answer

You should not write your name anywhere on your answers.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

[Examination instructions continue on next page...]

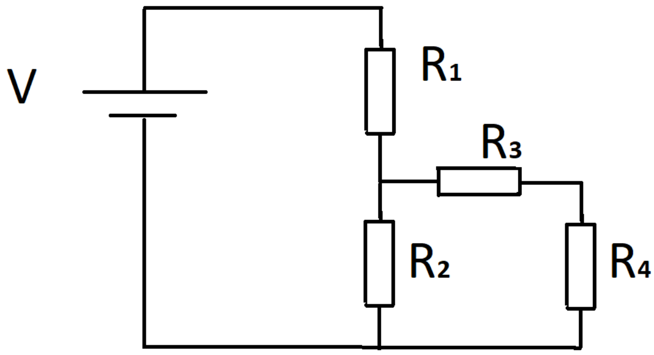
While this time-limited remote assessment has not been designed to be open book, in the present circumstances it is being run as an open-book examination. We have worked hard to create exams that assess synthesis of knowledge rather than factual recall. Thus, access to the internet, notes or other sources of factual information in the time provided will not be helpful and may well limit your time to successfully synthesise the answers required. Where individual questions rely more on factual recall and may therefore be less discriminating in an open book context, we may compare the performance on these questions to similar style questions in previous years and we may scale or ignore the marks associated with such questions or parts of the questions. The use of the work of another student, past or present, constitutes plagiarism. Giving your work to another student to use may also constitute an offence. Collusion is a form of plagiarism and will be treated in a similar manner. This is an individual assessment and thus should be completed solely by you. The College will investigate all instances where an examination or assessment offence is reported or suspected, using plagiarism software, vivas and other tools, and apply appropriate penalties to students. In all examinations we will analyse exam performance against previous performance and against data from previous years and use an evidence-based approach to maintain a fair and robust examination. As with all exams, the best strategy is to read the question carefully and answer as fully as possible, taking account of the time and number of marks available.

SECTION A

1. (i) (a) Briefly state Thévenin's theorem.

(b) In the circuit shown in the diagram below the resistor R_4 is removed. Express the Thévenin voltage between the resulting open ends of the circuit as a function of the voltage V and other resistances in the circuit.

(c) Find the expression for the Thévenin resistance and sketch the Thévenin equivalent circuit, assuming the resistor R_4 was removed in the diagram below.



(d) Briefly state Norton's theorem.

(e) Express the Norton current flowing through a wire replacing the removed resistor R_4 in the circuit shown in the diagram above.

(f) Give the expression for the Norton resistance and sketch the Norton equivalent circuit, assuming the resistor R_4 was removed in the diagram above.

(g) Calculate the numerical values of the Thévenin voltage, the Thévenin resistance, the Norton resistance and the Norton current assuming V=20 V, R_1=16 Ω, R_2=12 Ω, and R_3=10 Ω.

(h) Verify the equivalence of the Thévenin circuit and the Norton circuit representations of the diagram above, by comparing the values of the currents through the resistor R_4=5 Ω.

[20 marks]

(ii) (a) Briefly state the properties of the ideal operational amplifier and give the two 'golden-rules' used to simplify the analysis of circuits containing them.

(b) The diagram below shows a circuit containing an operational amplifier which is assumed to be ideal. Derive the equation for the current flowing

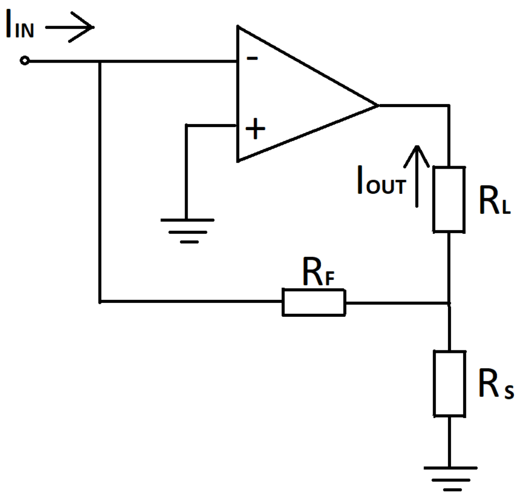
[This question continues on the next page . . .]

through the load resistor I_{OUT} as a function of the input current I_{IN} and the resistances of resistors in the circuit.

(c) Calculate the voltage drop across the load resistor R_L=8 Ω, assuming the input current I_{IN}=0.5 A and resististances of the other resistors to be R_F=20 Ω and R_S=10 Ω.

(d) Briefly state the properties of typical 'real-world' operational amplifiers.

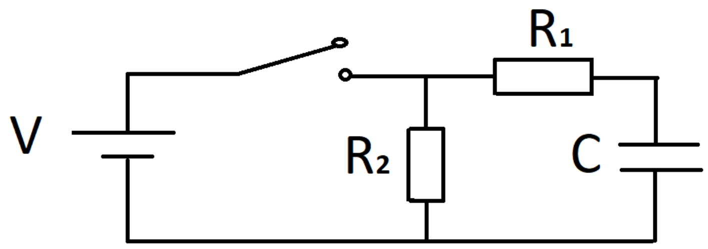
[20 marks]



[Total 40 marks]

SECTION B

2. (i) The RC circuit in the diagram below has a DC voltage applied and contains a switch, which is initially closed for a long time. The value of the source voltage is V=10 V, the value of capacitance is C=2 μF, and resistances of the two resistors are R_1=150 $k\Omega$ and R_2=300 $k\Omega$.



(a) Relate the voltage across the capacitor to the voltage of the DC source V before the switch is opened.

(b) Write the differential equation for the current flowing through the resistor R_2 after the switch is opened.

(c) Solve the differential equation for the current flowing through the resistor R_2 after the switch is opened.

(d) Plot the current flowing through the resistor R_2 after the switch is opened as a function of time.

(e) Calculate the voltage across the capacitor 1.5 s after the switch was opened.

[12 marks]

(ii) (a) Define the transfer function of an electronic system with respect to its input and output signals.

(b) Explain the importance of the impulse reponse of a system and its relation to the transfer function, using relevant equations.

[6 marks]

(iii) (a) Find the time-domain impulse-response of a system with a transfer function given by the equation below.

$$G(s) = \frac{4s + 8}{4s^3 + 16s^2 + 13s + 3}$$

(b) Find the zeros and poles of the transfer function $G(s)$ and comment on the stability of the system it represents.

[This question continues on the next page . . .]

(c) Sketch the time-domain impulse-response of the system.

[12 marks]

Hint: You may use the table of standard Laplace Transforms at the end of this paper. [Total 30 marks]

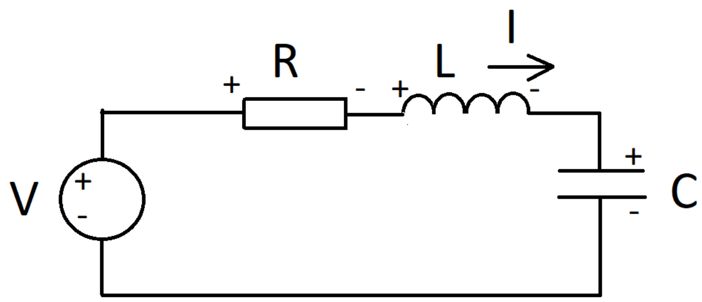
3. (i) (a) Define the Laplace transform $F(s)$ of the continuous function $f(t)$ and explain the difference with respect to the Fourier transform.

(b) Calculate the Laplace transform for the function $f(t)$ defined as:

$$f(t) = \begin{cases} 0, & t < 0 \\ t, & 0 \le t \le 1 \\ 0, & t > 1 \end{cases}$$

[7 marks]

(ii) Consider the LRC circuit shown below consisting of a voltage source, inductance of 1 H, capacitance of 1 F and resistance of 2 Ω. The voltage source produces the impulse $V = \delta(t)$. At time $t = 0$, the capacitor is charged and its initial voltage is -4 V, and the intital current through the inductor is 2 A.



(a) Write the equation describing the voltages in the circuit.

(b) Write the differential equation describing the current flowing in the circuit.

(c) Take the Laplace transform of the equation describing the current in the circuit and write its corresponding s-domain representation.

(d) By performing the inverse Laplace transform, find an expression for the current flowing in the circuit for time $t > 0$.

(e) Sketch the current flowing in the circuit for time $t > 0$ and calculate its value at $t = 2\,\text{s}$.

[23 marks]

Hint: You may use the table of standard Laplace Transforms at the end of this paper. [Total 30 marks]

4. (i) A photomultiplier tube (PMT) with 12 dynodes has an electron gain of 10^8.

(a) Calculate how many additional electrons are released, on average, at each dynode.

(b) Assuming that this PMT has a photocathode with quantum efficiency of 25% for light with 450 nm wavelength, calculate the average total charge created per individual photon of that wavelength entering the tube.

(c) The signal from the PMT is connected to the 50 Ω input of an oscilloscope. Estimate the measured voltage created by a single photoelectron, assuming that the average transit time spread through the PMT is about 6 ns (Full Width at Half Maximum - FWHM).

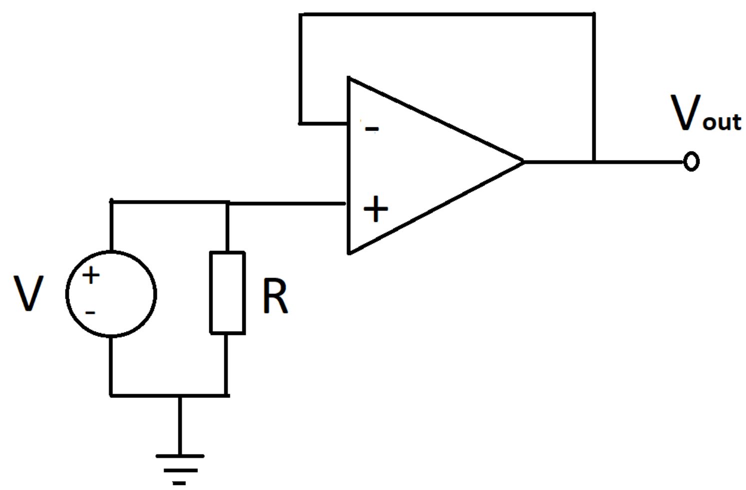
(d) Define the so called 'Dark Current' explaining its physical origin and its effect on measurements using PMTs.

[14 marks]

(ii) A voltage source of 1 V is connected to a 100 kΩ resistor. The voltage across the resistor is measured at a temperature of 300 K via the voltage follower as shown in the figure below, assuming that the noise contribution from the voltage source can be neglected. The voltage follower has a noise figure of $F = 4$, where the noise figure is defined as:

$$F = \left(\frac{P_S}{P_N}\right)_{IN} / \left(\frac{P_S}{P_N}\right)_{OUT}$$

with P_S (P_N) a signal (noise) power either at input or output, respectively.



(a) Calculate the thermal RMS voltage noise from the resistor at the ouput of the voltage follower measured with a bandwidth of 1 kHz.

(b) Calculate the RMS flicker noise voltage originating from the resistor assuming that it is based on technology in which the noise power between 3 kHz and 1 kHz equals 9.1×10^{-20} W. Assume that the measurement is performed using the circuit above at a centre frequency of 600 Hz and with the same bandwidth of 1 kHz.

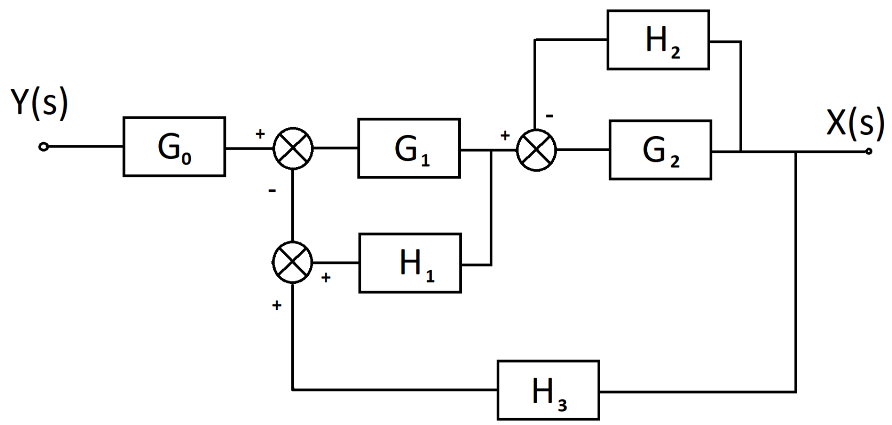
(c) What is the total voltage noise measured in the circuit above, taking into account the thermal and flicker noise contributions?

[12 marks]

(iii) Sketch a plot of the overall noise power versus frequency which we may expect to see in a typical measurement, labelling the contribution from all the important sources. [4 marks]

[Total 30 marks]

5. The system shown below contains six functional blocks and three feedback loops.



(i) (a) Calculate the system transfer function $G(s)$ in terms of $G_0(s)$, $G_1(s)$, $G_2(s)$, $H_1(s)$, $H_2(s)$ and $H_3(s)$.

(b) $G_0(s)$ has a time-domain impulse response of $2te^{-t}$. Find its s-domain representation.

(c) $G_1(s)$ is an ideal differentiator with unit gain. Find its s-domain representation.

(d) $G_2(s)$ is an ideal integrator with unit gain. Find its s-domain representation.

(e) $H_2(s)$ is an ideal amplifier with gain 6. Find is its s-domain representation.

(f) Find the overall transfer function in terms of s, assuming that $H_1(s)$ and $H_3(s)$ are ideal amplifiers with gains 1, and -2, respectively.

[15 marks]

(ii) (a) Is the system stable? Explain your answer.

(b) Would the circuit oscillate?

(c) Determine the system response to a unit step input in the time-domain.

[15 marks]

Hint: You may use the table of standard Laplace Transforms at the end of this paper. [Total 30 marks]

Table of Laplace Transforms

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
t^n for $n > 0$	$\frac{n!}{s^{(n+1)}}$
e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$
$\frac{1}{b-a}(e^{-at} - e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
$e^{-at}\sin(\omega t)$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at}\cos(\omega t)$	$\frac{s+a}{(s+a)^2+\omega^2}$